



Efficient Brain Imaging Analysis for Alzheimer's and Dementia Detection Using Convolution-Derivative Operations

Yasmine Mustafa*, Mohamed Elmahallawy**, Tie Luo†

*Computer Science Department, Missouri University of Science and Technology, Rolla, MO 65401, USA

**Department of Computer Science and Cybersecurity, Washington State University, Richland, WA 99354, USA

†Department of Electrical and Computer Engineering, University of Kentucky, Lexington, KY 40506, USA

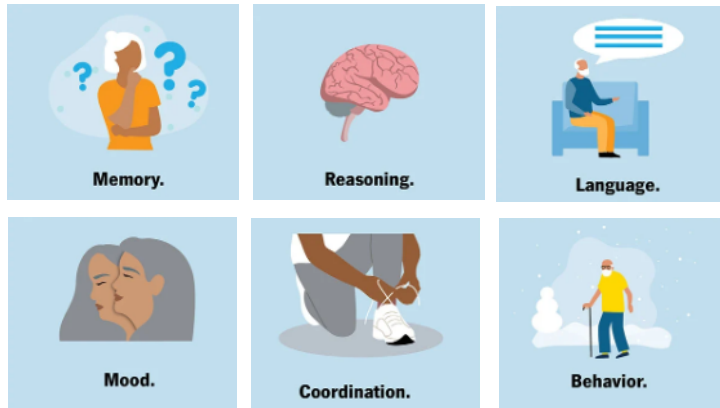
Alzheimer's Disease (AD)

- The lifetime risk for Alzheimer's disease at age 45:

1 in **5** women and **1** in **10** men.



- Kills more people than prostate cancer and breast cancer combined
- The number of people living with AD **doubles** every 5 years, estimated to reach **14 million** by **2060**.
- The symptoms of AD involve a gradual decline in some, most or all of the following:



Detecting Atrophy

Challenges

- Difficult to detect subtle changes
- Traditionally relies on visual assessments.
- Manual measurements by radiologists are time-consuming.



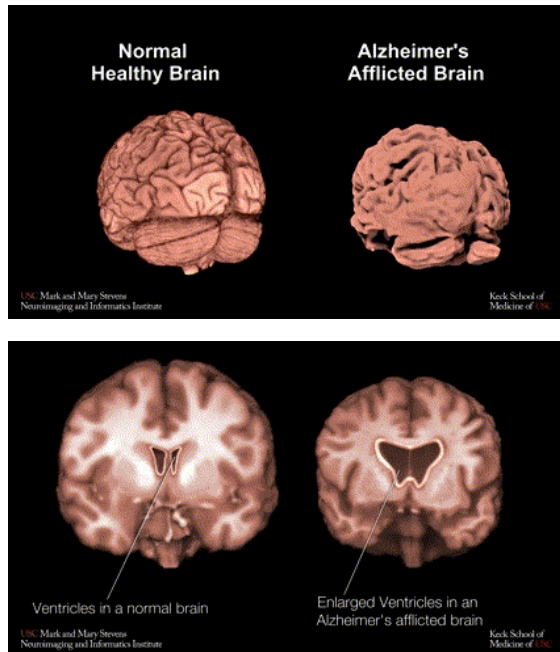
Advancements

- Computer-Aided Diagnostic (CAD) tools.
- Integration of advanced imaging techniques and ML.

Voxel-Based Morphometry (VBM)

- Statistical analysis across 3D voxels.
- Detects brain changes due to atrophy or tissue expansion.
- Highlights anatomical differences between groups.

Brain Volume Changes Denoting Dementia



Jacobian Maps

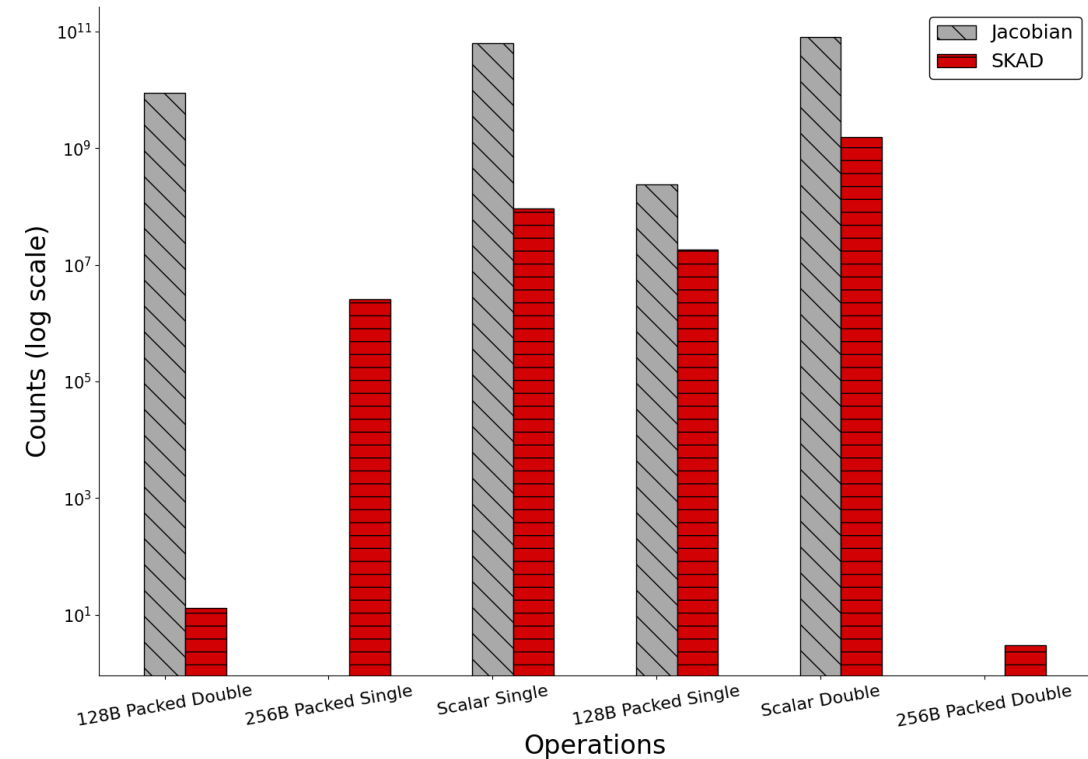
- Capture subtle **brain volume changes**
- Providing informative representations for feature learning
- Highlight **local brain morphometry**

$$J(v) = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{bmatrix}$$

SKAD: Sobel Kernel Angle Difference

Efficient Analysis for Neurodegenerative Disorders

- **Efficient and Accurate:**
 - Analyzes volumetric changes in Alzheimer's, dementia, and Parkinson's.
 - **6x faster** than Jacobian maps while maintaining accuracy.
- **Computational Gains:**
 - **Time Reduction:** From 584 ms $\rightarrow$ 93 ms (**84% faster**).
 - **FLOPs Reduction:** From 146B $\rightarrow$ 1.66B (**99% less**).



SKAD significantly outperforms Jacobian maps in computational efficiency.

Datasets, Labeling, and Balancing



- Participants include **755 cognitively normal adults** and **622 individuals at various stages** of cognitive decline ranging in **age from 42-95 yrs.**



- Participants include **755 cognitively normal adults** and **622 individuals at various stages** of cognitive decline ranging in **age from 40-89 yrs.**

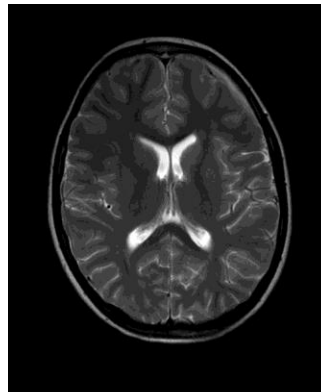
- Based on **clinical dementia rating (CDR)** scores:

CDR	Class
0	Normal
0.5	MCI
1	Mild
2	Moderate
3	Severe

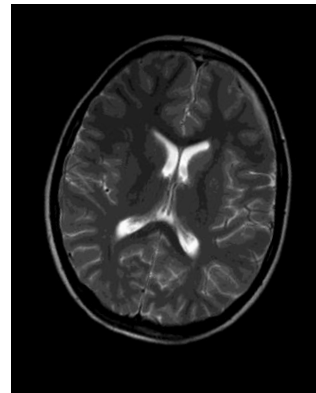
} **Combined**

Datasets, Labeling, and Balancing

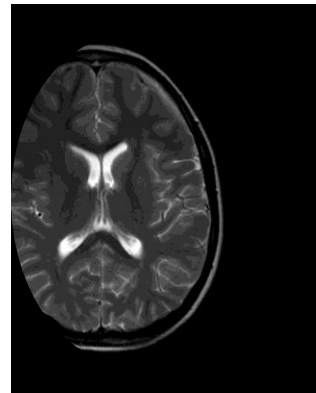
Data Augmentation: Illustration from Axial View



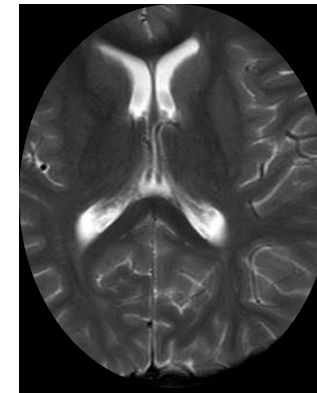
Original Image



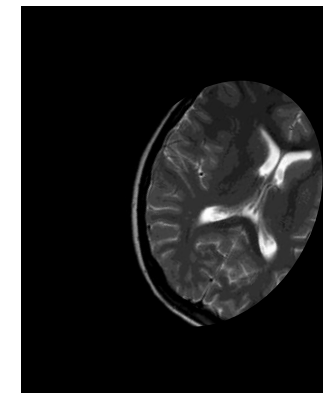
Rotate:
 $\pm 10^\circ$



Translate:
 ± 5

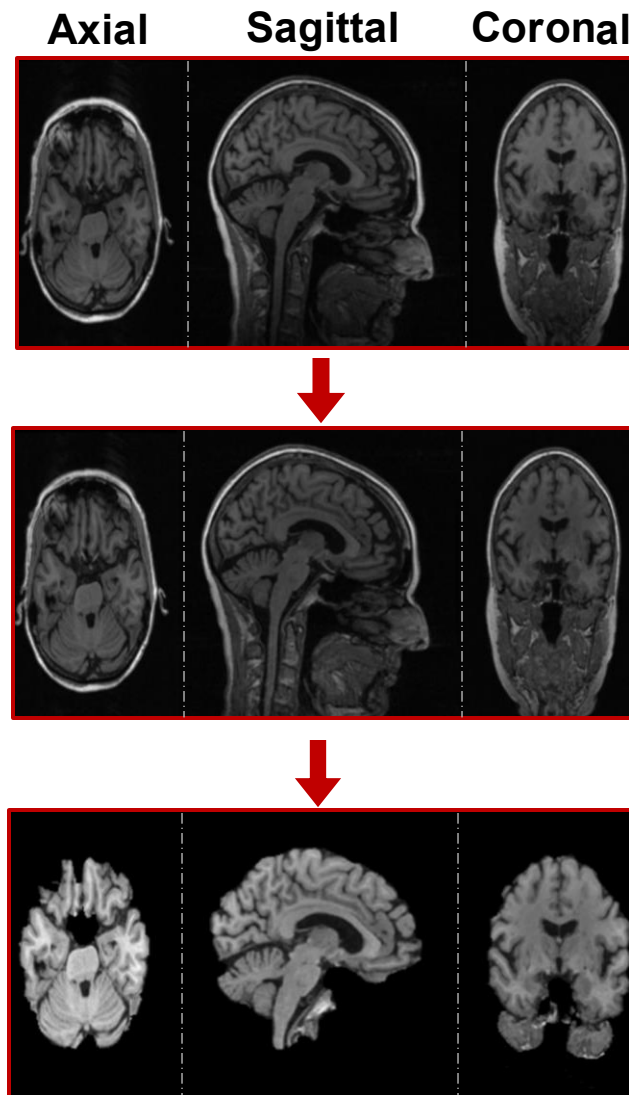


Zoom in:
[1.1 - 1.3]



Rototranslate

Data Preprocessing and Registration



1. **Reorientation:** All images are reoriented to align with a standard template.

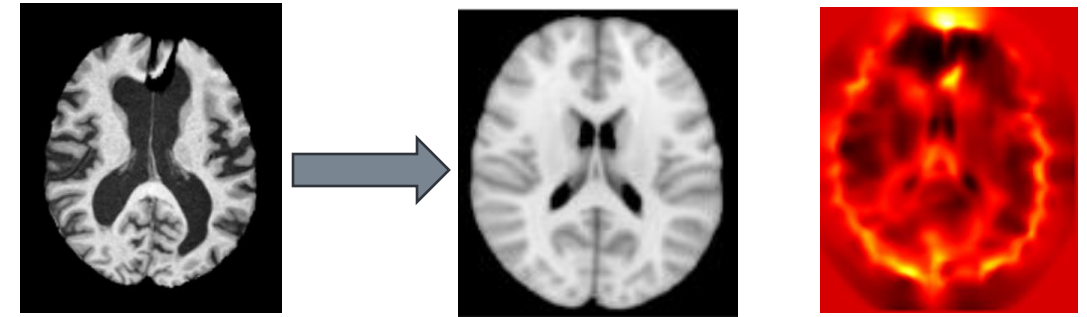
2. **Bias Field Correction:** Signal intensities across the image are unified

3. **Brain extraction:** Non-brain tissues are removed from the MRI

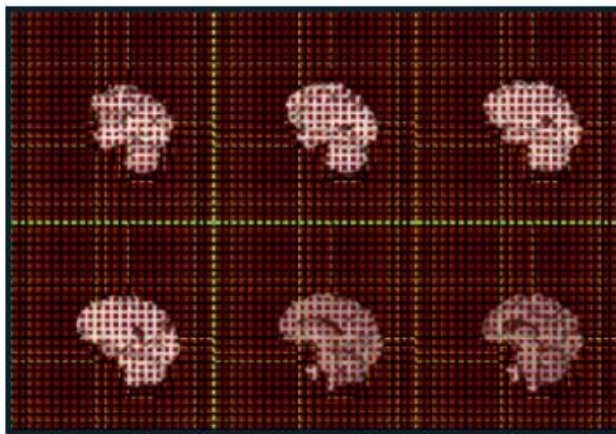
Data Preprocessing and Registration

Purpose of Registration

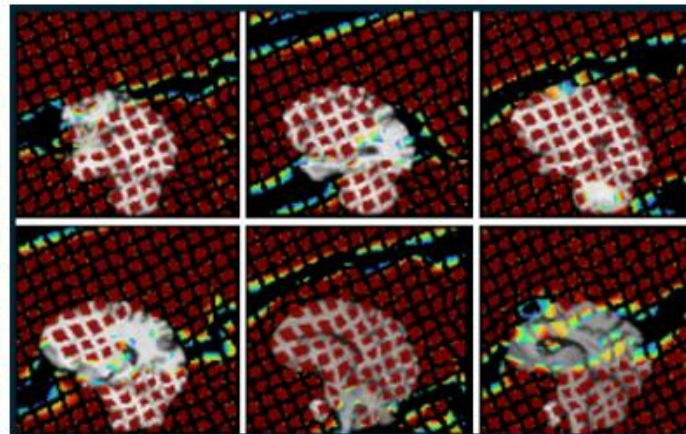
- Spatial alignment with a healthy template for comparative analysis.
- Standardization to assess deviations in brain morphology.
- Highlights volumetric changes via warped grids.
- **Deformation Field (V):** Maps voxel displacements for precise matching.



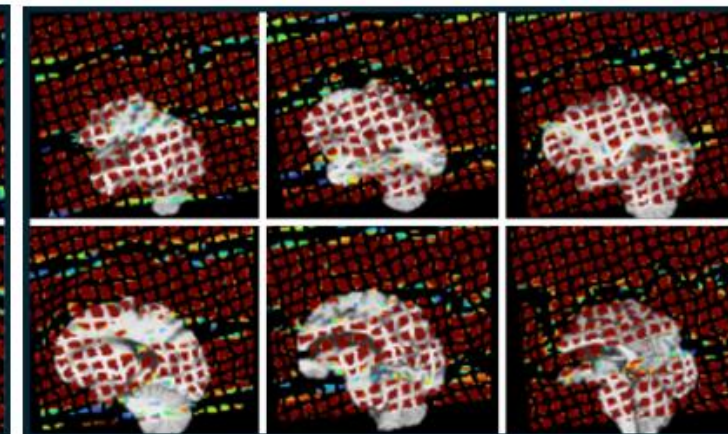
Transform all MRI to a
standard common space
MNI152



(a) Original grid.



(b) Warped grid of a cognitively normal subject.



(c) Warped grid of a subject with dementia.

Jacobian Maps



Transform all MRI to a
standard common space
MNI152

What is a Jacobian Map?

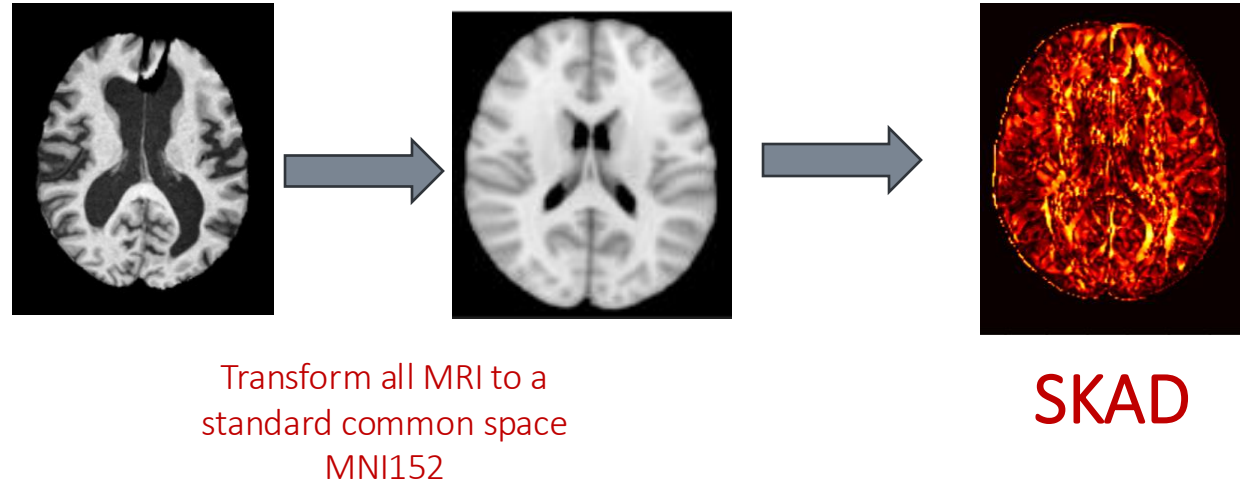
- A matrix J derived from the deformation field V .
- Encodes local deformations through first derivatives of voxel positions.
- Identifies voxel-level volumetric changes before and after deformation.

$$J(v) = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{bmatrix}$$

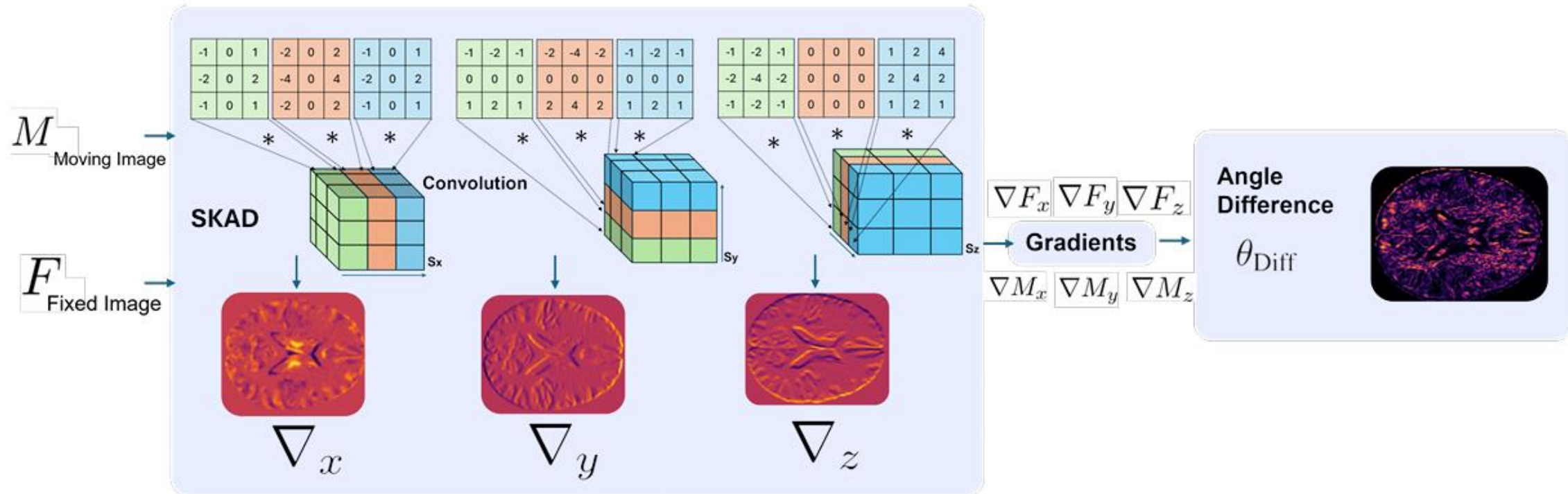
$$J_{map}(M) = \begin{bmatrix} \vdots & & \\ \dots & Det(J(v(x, y, z))) & \dots \\ \vdots & & \end{bmatrix} \begin{matrix} x = 1 \dots W \\ y = 1 \dots H \\ z = 1 \dots D \end{matrix}$$

$$\text{At each voxel: } \begin{cases} \text{volume expansion} & \text{if } Det(J) > 1 \\ \text{no change} & \text{if } Det(J) = 1 \\ \text{volume compression} & \text{if } Det(J) < 1 \end{cases}$$

Jacobian Maps



Jacobian Maps



$$\theta_{\text{Diff}} = \left| \tan^{-1} \left(\frac{\sqrt{\nabla M_x^2 + \nabla M_y^2}}{\nabla M_z} \right) - \tan^{-1} \left(\frac{\sqrt{\nabla F_x^2 + \nabla F_y^2}}{\nabla F_z} \right) \right|$$

Computational Complexity and Accuracy

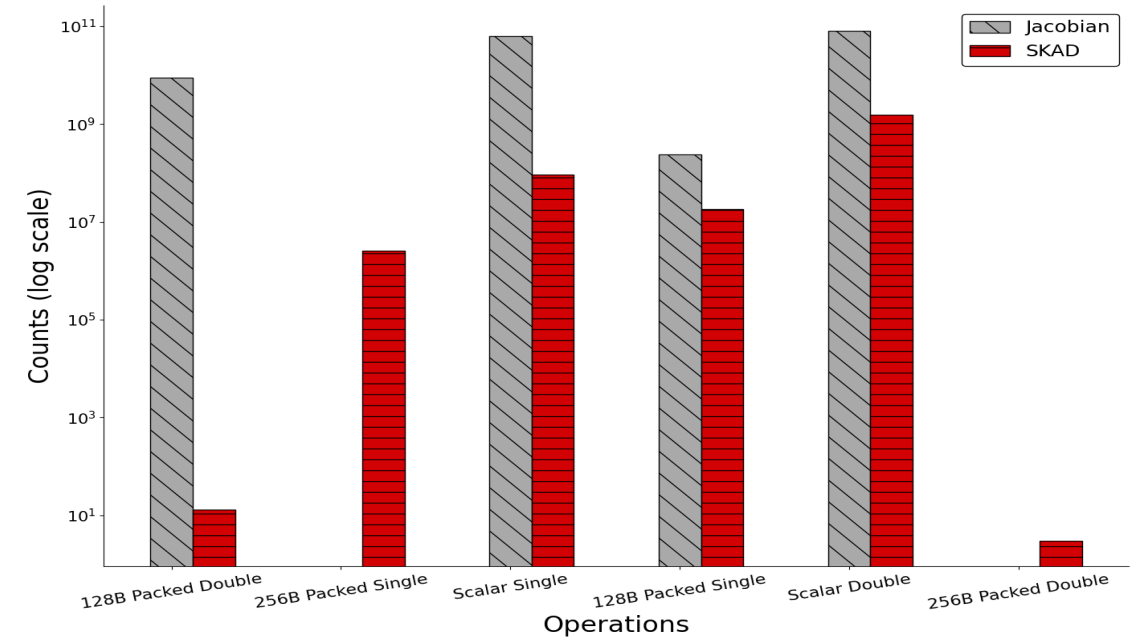
Data	Input Type	Sensitivity (%)					Specificity (%)					Accuracy (%)				
		Avg	CN	MCI	MLD	SEV	Avg	CN	MCI	MLD	SEV	Avg	CN	MCI	MLD	SEV
ADNI	Registered	74.4	68.5	77.2	69.9	82.0	91.4	89.4	89.3	93.8	93.1	87.1	84.0	86.2	89.3	88.9
	Jacobian	87.8	80.3	75.1	96.7	99.0	96.0	94.7	92.2	97.7	99.4	94.0	90.7	88.0	97.8	99.6
	SKAD	83.5	78.10	88.6	78.42	89.04	94.3	91.65	89.86	95.80	100.0	91.06	88.65	84.23	91.94	99.42
OASIS	Registered	77.7	37.5	94.2	81.3	97.9	92.4	98.7	74.1	97.9	99.0	88.7	82.9	78.3	94.2	99.6
	Jacobian	83.9	64.4	84.4	87.8	99.0	92.8	85.7	98.6	87.8	99.0	89.4	85.0	85.0	87.9	99.8
	SKAD	84.6	67.26	84.13	94.54	92.67	93.9	94.46	98.50	82.99	99.64	94.3	92.35	94.43	91.64	98.88

Function	Total Time (ms)	Memory Usage (MiB)
Jacobian Map	584	1267.3
SKAD	93	1199.7

Discussion and Conclusion

Why is SKAD More Efficient?

- **Simplified Calculations:**
 - Uses Sobel kernels instead of complex derivative computations.
- **Dimensionality Reduction:**
 - Single angle per voxel vs. 3x3 matrix for Jacobian maps.
- **Optimized Convolutions:**
 - Efficiently supported on modern GPUs and libraries.
- **Single-Precision Operations:**
 - Reduces reliance on costly double-precision calculations.



Discussion and Conclusion

Key Takeaways

- Efficient alternative to Jacobian maps for detecting brain volumetric changes.
- Comparable diagnostic performance with **84% faster processing** and **99% lower FP operations**.
- Promotes scalable and accessible dementia diagnostics.
- Potential applicability to diverse datasets (e.g., NACC, HCP).

Future Work

- **Adaptive Sobel Kernels:**
Dynamically adjust size and orientation for finer gradient detection.
- **Explanatory Mechanisms:**
Add post-segmentation interpretation and brain region labeling.
- **Broader Applications:**
Extend to other neurodegenerative diseases, such as Parkinson's.

Thank You

